Roll No. | | | | | | | |

END SEMESTER EXAMINATION 2025

Name of the Course: M.TechSemester: II

Name of the Paper: Finite Element Analysis PaperCode:**MME 201**

Time: 3HoursMaximum Marks: 100

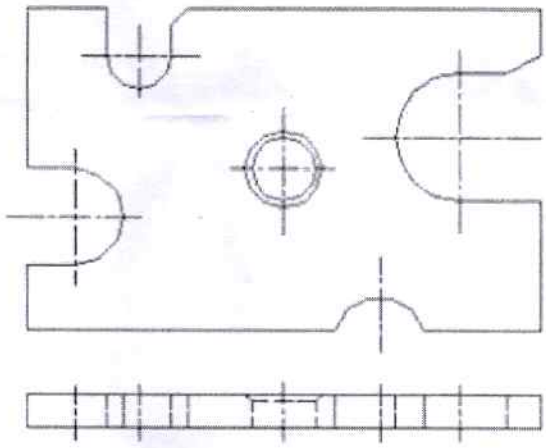
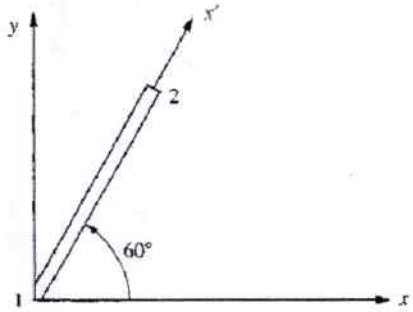
Note:-

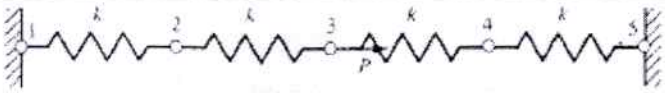
(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20marks)	**CO1,CO2**
(a)	Explain processing and post processing?	
(b)	Why finite element method may not be suitable for heat flow problems?	
(c)	How can you mesh this machine part in 10 elements. Explain why you did so? 	
Q2	(20 marks)	**CO2, CO3**
(a)	Derive the stiffness matrix for threeparallelly connected spring elements.	
(b)	What are degrees of freedom of a beam element? Why a cubic polynomial a good choice for modelling transverse displacement of a beam element?	
(c)	Derive the stiffness matrix for two connected beam elements.	
Q3	(20 marks)	**CO2, CO4**
(a)	Derive the global stiffness matrix for a bar element in 2D and evaluate it for the bar element shown in the figure below. E = 190 GPa, A = $2x10^4 m^2$, L = 1m. 	

(b)	 Determine nodal displacements, stress in elements and reaction force.	
(c)	Derive the transformation matrix for transforming a vector from local to global coordinates in 2D such that the local coordinates are rotated with respect to the global coordinates by an angle theta.	
Q4	(20 marks)	**CO4, CO5**
(a)	Derive the shape functions for a beam element.	
(b)	Derive the element stiffness matrix for beam element.	
(c)	How can two beam elements be combined to form a simply supported beam having a point load in the middle. Derive the global stiffness matrix.	
Q5	(20 marks)	**CO5, CO6**
(a)	How can the finite element method be applied to an oscillating system?	
(b)	Derive the finite element formulation of a spring mass system with two equal masses connected in series with springs of equal stiffness.	
(c)	Draw, derive and explain the shape functions for a bar element.	

Note for the question paper setters:

- **Question paper should cover all the COs of the course.**